EXTENSION 2

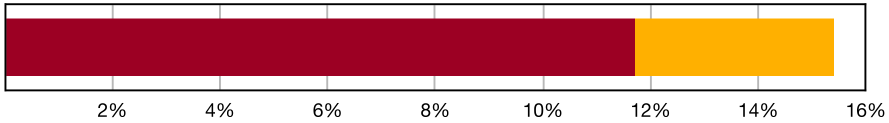
Complex Numbers (Ext2), N2 Using Complex Numbers (Ext2)

Geometrical Implications of Complex Numbers

Teacher: Nardene Dodd

Exam Equivalent Time: 159 minutes *(based on allocation of 1.5 minutes per mark)*

N2 Using Complex Numbers



Solving Equations
Geometrical Implications

***SmarterMaths analytics based on the average contribution to Extension 2 exams since 2020.**

HISTORICAL CONTRIBUTION

- *N2 Using Complex Numbers* has contributed a substantial 15.3% per new syllabus Ext2 exam since it was introduced in 2020.
- We have split this topic into two categories for analysis purposes which are: *1-Solving Equations With Complex Numbers (11.6%) and 2-Geometrical Implications of Complex Numbers (3.7%).*
- This analysis looks at *Geometrical Implications of Complex Numbers.*

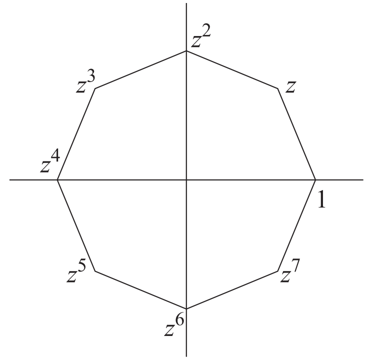
HSC ANALYSIS - What to expect and common pitfalls

- *Geometrical Implications of Complex Numbers (3.7%)* has appeared in the longer answer section of each new syllabus exam in questions worth 3-5 marks.
- An important take home in looking at this topic area is the high level of difficulty of the associated questions, with mean marks of 21% and 15% produced by *2022 Ext2 16a* and *2021 Ext2 16c* respectively.
- Sketching regions has been examined twice in new syllabus exams with *2022 Ext2 1 MC* demonstrating the band 3 difficulty level while *2021 Ext2 16c* tested students at the high band 6 level.
- Exponential form and the geometrical rotation of associated complex numbers featured in *2020 Ext2 14a*. This question also presented a challenging proof and should appear in any revision set.
- We note this topic area now has to compete with *Vectors* for mark allocations in the new syllabus and its contributions will suffer from this in our view. *2021 Ext2 10 MC* combined both vectors and complex numbers in a high difficulty question deserving attention.

Questions

1. Complex Numbers, EXT2 N2 2017 HSC 1 MC

The complex number z is chosen so that $1, z, \dots, z^7$ form the vertices of the regular polygon shown.



Which polynomial equation has all of these complex numbers as roots?

A. $x^7 - 1 = 0$

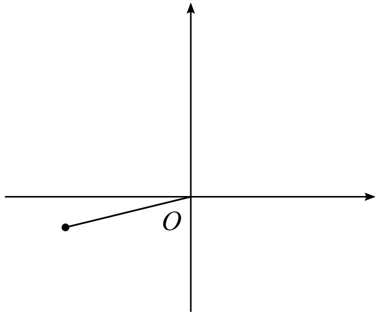
B. $x^7 + 1 = 0$

C. $x^8 - 1 = 0$

D. $x^8 + 1 = 0$

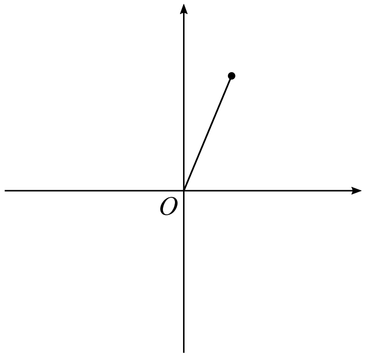
2. Complex Numbers, EXT2 N2 EQ-Bank 1 MC

The Argand diagram shows the complex number $e^{i\theta}$.

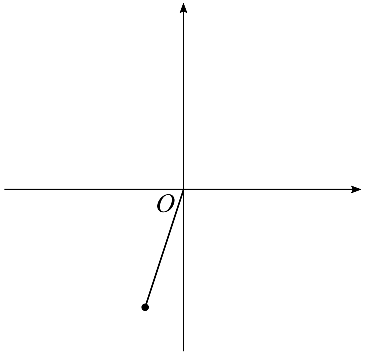


Which of the following could be the complex number $ie^{-i\theta}$?

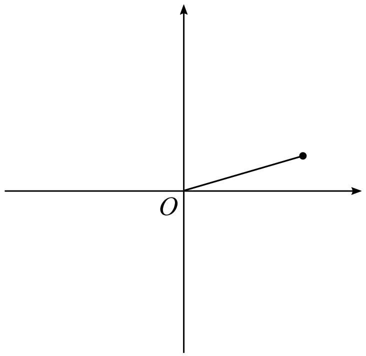
A.



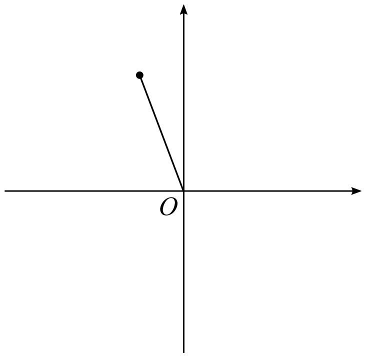
B.



C.



D.



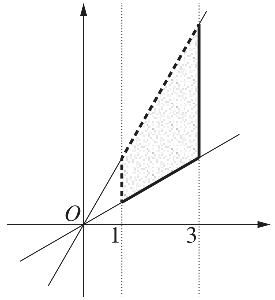
---

3. Complex Numbers, EXT2 N2 2022 HSC 1 MC

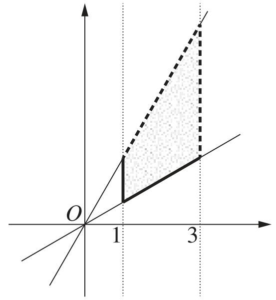
Let R be the region in the complex plane defined by $1 < \mathrm{Re}(z) \leq 3$ and $\frac{\pi}{6} \leq \mathrm{Arg}(z) < \frac{\pi}{3}$.

Which diagram best represents the region R?

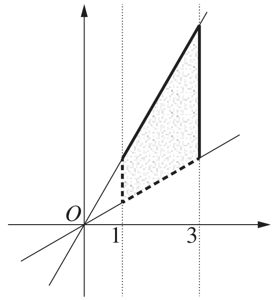
A.



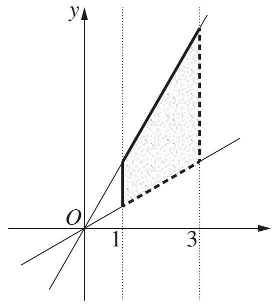
B.



C.



D.



---

4. Complex Numbers, EXT2 N2 2017 HSC 3 MC

Which complex number lies in the region $2 < |z-1| < 3$?

A. $1+\sqrt{3}i$

B. $1+3i$

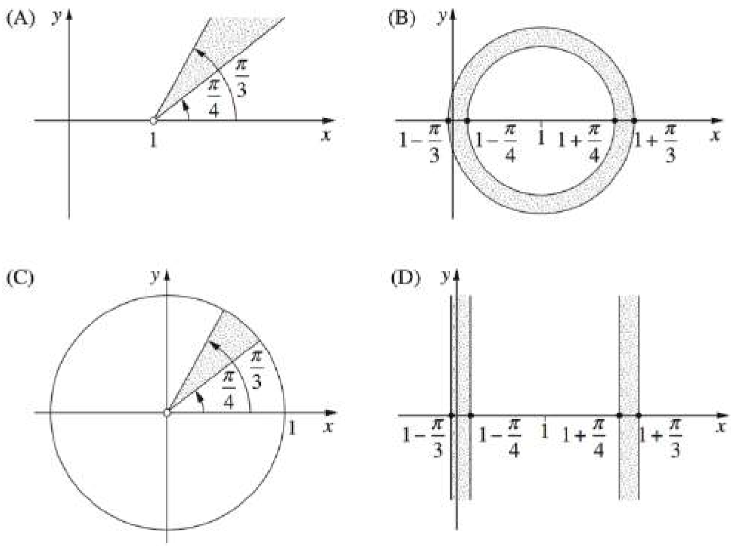
C. $2+i$

D. $3-i$

---

5. Complex Numbers, EXT2 N2 2013 HSC 5 MC

Which region on the Argand diagram is defined by $\frac{\pi}{4} \leq |z-1| \leq \frac{\pi}{3}$?



---

6. Complex Numbers, EXT2 N2 2015 HSC 9 MC

The complex number z satisfies $|z-1|=1.$

What is the greatest distance that z can be from the point i on the Argand diagram?

A. 1

B. $\sqrt{5}$

C. $2\sqrt{2}$

D. $\sqrt{2}+1$

---

7. Complex Numbers, EXT2 N2 2016 HSC 5 MC

Multiplying a non-zero complex number by $\frac{1-i}{1+i}$ results in a rotation about the origin on an Argand diagram.

What is the rotation?

A. Clockwise by $\frac{\pi}{4}$

B. Clockwise by $\frac{\pi}{2}$

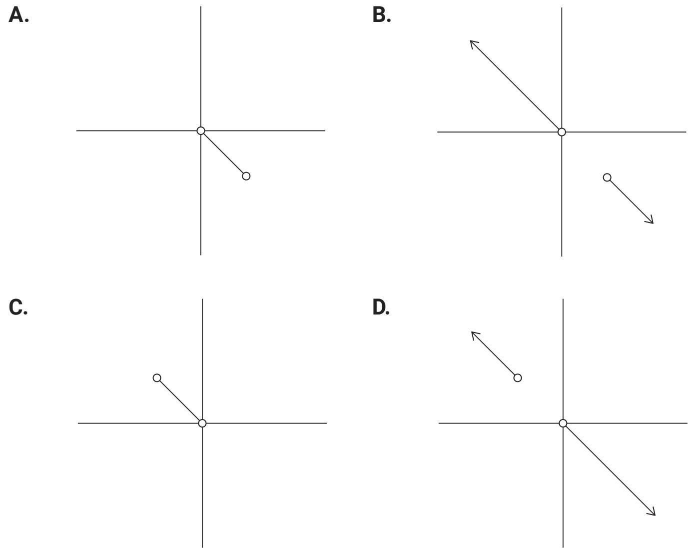
C. Anticlockwise by $\frac{\pi}{4}$

D. Anticlockwise by $\frac{\pi}{2}$

---

8. Complex Numbers, EXT2 N2 2018 HSC 7 MC

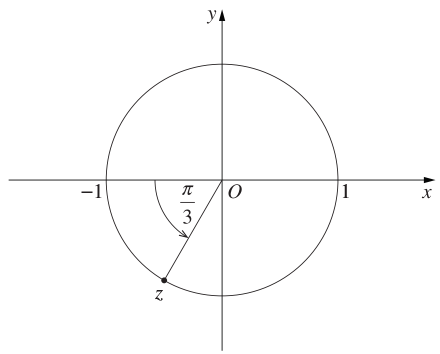
Which diagram best represents the solutions to the equation $\arg(z) = \arg(z+1-i)$?



---

9. Complex Numbers, EXT2 N2 2023 HSC 3 MC

A complex number z lies on the unit circle in the complex plane, as shown in the diagram.



Which of the following complex numbers is equal to $\bar{z}$?

A. $-z$

B. z^2

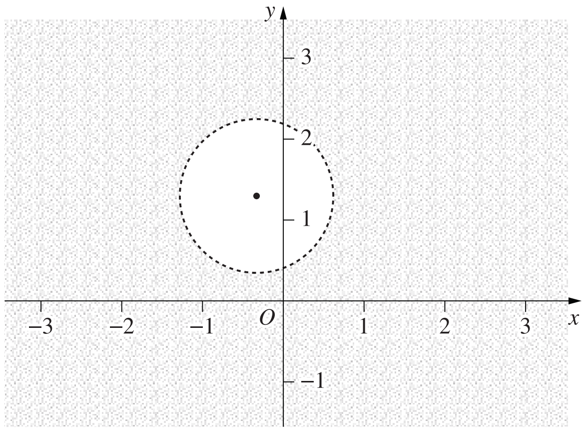
C. $-z^3$

D. z^4

---

10. Complex Numbers, EXT2 N2 2023 HSC 8 MC

A shaded region on a complex plane is shown.



Which relation best describes the region shaded on the complex plane?

A. $|z-i| > 2|z-1|$

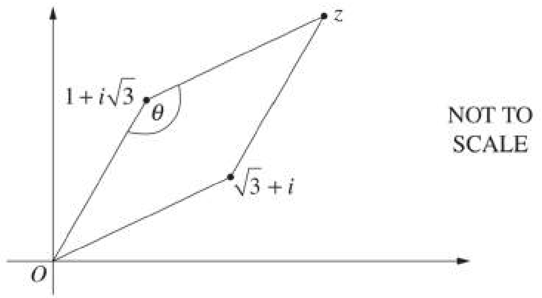
B. $|z-i| < 2|z-1|$

C. $|z-1| > 2|z-i|$

D. $|z-1| < 2|z-i|$

11. Complex Numbers, EXT2 N2 2011 HSC 2b

On the Argand diagram, the complex numbers $0, 1+i\sqrt{3}, \sqrt{3}+i$ and z form a rhombus.



i. Find z in the form $a+ib$, where a and b are real numbers. ***(1 mark)***

ii. An interior angle, θ, of the rhombus is marked on the diagram.
Find the value of θ. ***(2 marks)***

12. Complex Numbers, EXT2 N2 2009 HSC 2d

Sketch the region in the complex plane where the inequalities $|z-1| \leq 2$ and $-\frac{\pi}{4} \leq \arg(z-1) \leq \frac{\pi}{4}$ hold simultaneously. ***(2 marks)***

13. Complex Numbers, EXT2 N2 2010 HSC 2c

Sketch the region in the complex plane where the inequalities $1 \leq |z| \leq 2$ and $0 \leq z+\bar{z} \leq 3$ hold simultaneously. ***(2 marks)***

14. Complex Numbers, EXT2 N2 2012 HSC 11b

Shade the region on the Argand diagram where the two inequalities

$|z+2| \geq 2$ and $|z-i| \leq 1$

both hold. ***(2 marks)***

15. Complex Numbers, EXT2 N2 2017 HSC 11c

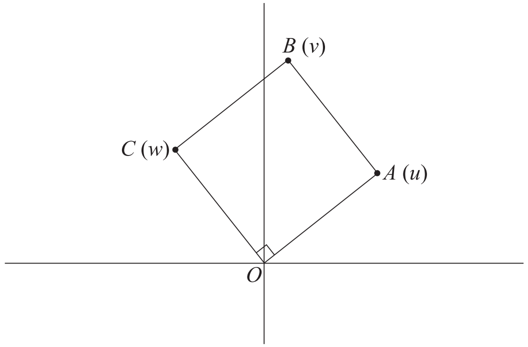
Sketch the region in the Argand diagram where

$-\frac{\pi}{4} \leq \arg(z) \leq 0$ and $|z-1+i| \leq 1$. ***(2 marks)***

16. Complex Numbers, EXT2 N2 2018 HSC 11d

The points A, B and C on the Argand diagram represent the complex numbers u, v and w respectively.

The points O, A, B and C form a square as shown on the diagram.



It is given that $u = 5 + 2i$.

i. Find w. *(1 mark)*

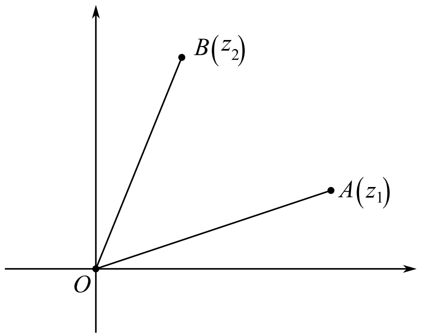
ii. Find v. *(1 mark)*

iii. Find $\arg\left(\dfrac{w}{v}\right)$. *(1 mark)*

17. Complex Numbers, EXT2 N2 2020 HSC 14a

Let z_1 be a complex number and let $z_2 = e^{\frac{i\pi}{3}} z_1$

The diagram shows points A and B which represent z_1 and z_2, respectively, in the Argand plane.



i. Explain why triangle OAB is an equilateral triangle. *(2 marks)*

ii. Prove that $z_1^2 + z_2^2 = z_1 z_2$. *(3 marks)*

18. Complex Numbers, EXT2 N2 2007 HSC 2c

The point P on the Argand diagram represents the complex number z, where z satisfies

$$\frac{1}{z} + \frac{1}{\overline{z}} = 1.$$

Give a geometrical description of the locus of P as z varies. *(3 marks)*

19. Complex Numbers, EXT2 N2 2013 HSC 11e

Sketch the region on the Argand diagram defined by $z^2 + \overline{z}^2 \leq 8$. *(3 marks)*

20. Complex Numbers, EXT2 N2 2014 HSC 11c

Sketch the region in the Argand diagram where $|\,z\,| \leq |\,z - 2\,|$ and $-\dfrac{\pi}{4} \leq \arg z \leq \dfrac{\pi}{4}$. *(3 marks)*

21. Complex Numbers, EXT2 N2 2019 HSC 12a

Sketch the region defined by $\dfrac{\pi}{4} \leq \arg(z) \leq \dfrac{\pi}{2}$ and $\operatorname{Im}(z) \leq 1$. *(2 marks)*

22. Complex Numbers, EXT2 N2 2004 HSC 2c

Sketch the region in the complex plane where the inequalities

$|z + \overline{z}| \leq 1$ and $|z - i| \leq 1$

hold simultaneously. *(3 marks)*

23. Complex Numbers, EXT2 N2 2005 HSC 2c

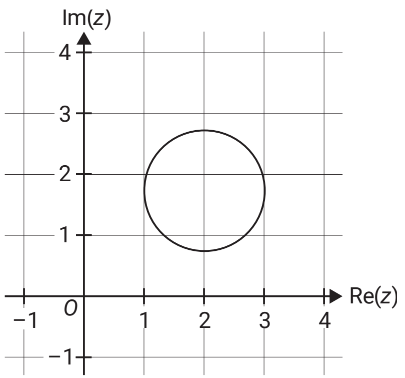
Sketch the region on the Argand diagram where the inequalities

$|z - \overline{z}| < 2$ and $|z - 1| \geq 1$

hold simultaneously. *(3 marks)*

24. Complex Numbers, EXT2 N2 2021 SPEC2 5

The graph of the circle given by $\left|z-2-\sqrt{3}i\right|=1$, where $z \in C$, is shown below.

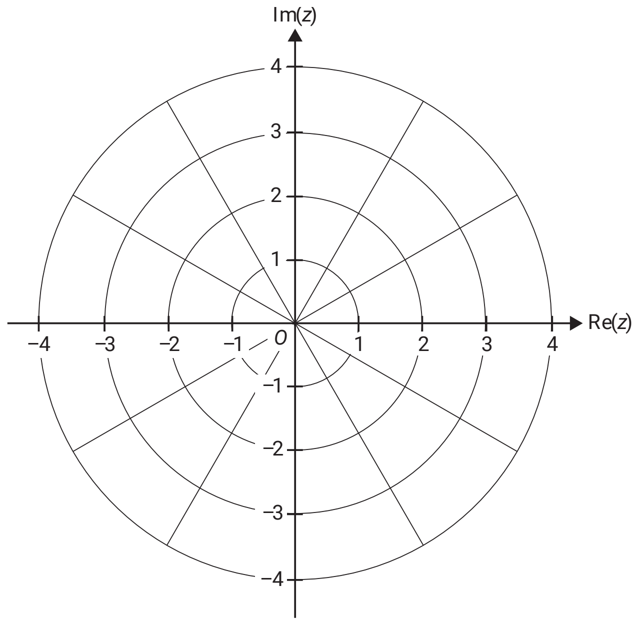


For points on this circle, find the maximum value of $|z|$. ***(2 marks)***

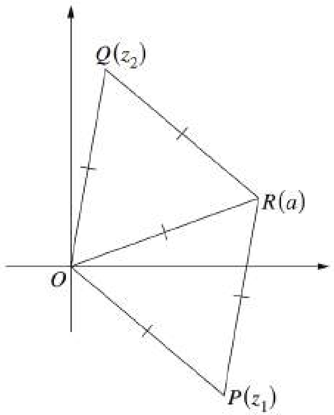
25. Complex Numbers, EXT2 N2 SM-Bank 11

Consider the point on the complex plane $z_1=\sqrt{3}+1$.

Sketch the ray given by $\operatorname{Arg}(z-z_1)=\dfrac{5\pi}{6}$ on the Argand diagram below. ***(2 marks)***



26. Complex Numbers, EXT2 N2 2007 HSC 2d



The points P, Q and R on the Argand diagram represent the complex numbers z_1, z_2 and a respectively.

The triangles OPR and OQR are equilateral with unit sides, so $|\,z_1\,|=|\,z_2\,|=|\,a\,|=1.$

Let $\omega=\cos\dfrac{\pi}{3}+i\sin\dfrac{\pi}{3}$.

i. Explain why $z_2=\omega a$. ***(1 mark)***

ii. Show that $z_1 z_2=a^2$. ***(1 mark)***

iii. Show that z_1 and z_2 are the roots of $z^2-az+a^2=0$. ***(2 marks)***

27. Complex Numbers, EXT2 N2 2023 14a

Let z be the complex number $z=e^{\frac{i\pi}{6}}$ and w be the complex number $w=e^{\frac{3i\pi}{4}}$.

i. By first writing z and w in Cartesian form, or otherwise, show that

$$|z+w|^2=\frac{4-\sqrt{6}+\sqrt{2}}{2}.$$ ***(3 marks)***

ii. The complex numbers z, w and $z+w$ are represented in the complex plane by the vectors $\overrightarrow{OA}, \overrightarrow{OB}$ and $\overrightarrow{OC}$ respectively, where O is the origin.

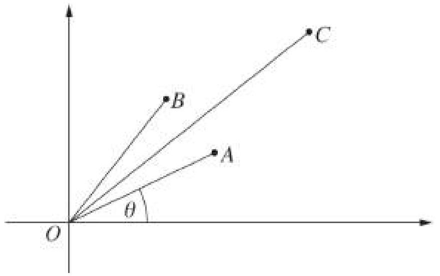
Show that $\angle AOC=\dfrac{7\pi}{24}$. ***(2 marks)***

iii. Deduce that $\cos\dfrac{7\pi}{24}=\dfrac{\sqrt{8-2\sqrt{6}+2\sqrt{2}}}{4}$. ***(1 mark)***

28. Complex Numbers, EXT2 N2 2010 HSC 2d

Let $z = \cos\theta + i\sin\theta$ where $0 < \theta < \dfrac{\pi}{2}$.

On the Argand diagram the point A represents z, the point B represents z^2 and the point C represents $z + z^2$.



Copy or trace the diagram into your writing booklet.

i. Explain why the parallelogram $OACB$ is a rhombus. *(1 mark)*

ii. Show that $\arg\left(z + z^2\right) = \dfrac{3\theta}{2}$. *(1 mark)*

iii. Show that $\left|z + z^2\right| = 2\cos\dfrac{\theta}{2}$. *(2 marks)*

iv. By considering the real part of $z + z^2$, or otherwise deduce that

$$\cos\theta + \cos 2\theta = 2\cos\frac{\theta}{2}\cos\frac{3\theta}{2}.$$ *(1 mark)*

29. Complex Numbers, EXT2 N2 2011 HSC 4a

Let a and b be real numbers with $a \neq b$. Let $z = x + iy$ be a complex number such that

$$|\, z - a\,|^2 - |\, z - b\,|^2 = 1.$$

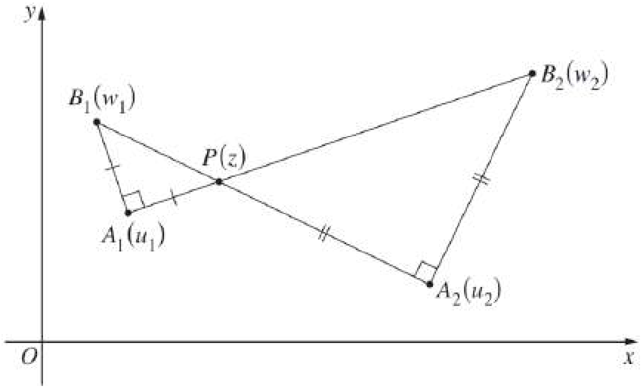
i. Prove that $x = \dfrac{a+b}{2} + \dfrac{1}{2(b-a)}$. *(2 marks)*

ii. Hence, describe the locus of all complex numbers z such that $|\, z - a\,|^2 - |\, z - b\,|^2 = 1.$ *(1 mark)*

30. Complex Numbers, EXT2 N2 2012 HSC 12d

On the Argand diagram the points A_1 and A_2 correspond to the distinct complex numbers u_1 and u_2 respectively. Let P be a point corresponding to a third complex number z.

Points B_1 and B_2 are positioned so that ΔA_1PB_1 and ΔA_2B_2P, labelled in an anti-clockwise direction, are right-angled and isosceles with right angles at A_1 and A_2, respectively. The complex numbers w_1 and w_2 correspond to B_1 and B_2, respectively.



i. Explain why $w_1 = u_1 + i(z - u_1)$. *(1 mark)*

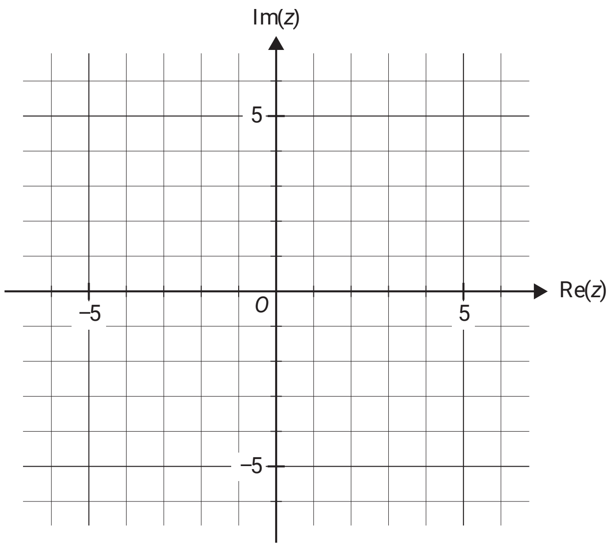
ii. Find the locus of the midpoint of B_1B_2 as P varies. *(2 marks)*

31. Complex Numbers, EXT2 N2 2020 SPEC2 2

Two complex numbers, u and v, are defined as $u = -2 - i$ and $v = -4 - 3i$.

a. Express the relation $|z-u| = |z-v|$ in the cartesian form $y = mx + c$, where $m, c \in R$. *(3 marks)*

b. Plot the points that represent u and v and the relation $|z-u| = |z-v|$ on the Argand diagram below. *(2 marks)*



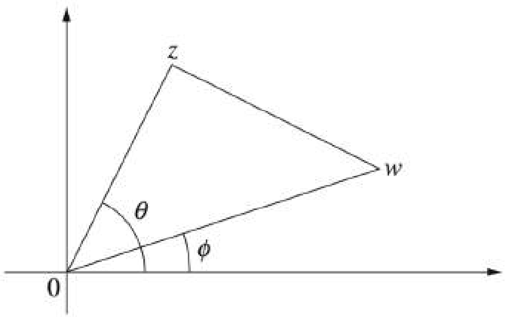
c. State a geometrical interpretation of the graph of $|z-u| = |z-v|$ in relation to the points that represent u and v. *(1 mark)*

d. **i.** Sketch the ray given by $\text{Arg}(z-u) = \dfrac{\pi}{4}$ on the Argand diagram in **part b.** *(1 mark)*

ii. In Cartesian form, write down the function that describes the ray $\text{Arg}(z-u) = \dfrac{\pi}{4}$. *(1 mark)*

32. Complex Numbers, EXT2 N2 2013 HSC 15a

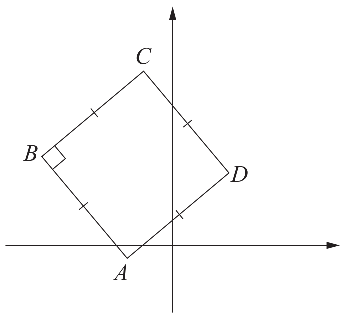
The Argand diagram shows complex numbers w and z with arguments ϕ and θ respectively, where $\phi < \theta$. The area of the triangle formed by $0, w$ and z is A.



Show that $z\overline{w} - w\overline{z} = 4iA$. *(3 marks)*

33. Complex Numbers, EXT2 N2 2017 HSC 13e

The points A, B, C and D on the Argand diagram represent the complex numbers a, b, c and d respectively. The points form a square as shown on the diagram.



By using vectors, or otherwise, show that $c = (1+i)d - ia$. *(2 marks)*

34. Complex Numbers, EXT2 N2 2016 HSC 16b

i. The complex numbers $0,\ u$ and v form the vertices of an equilateral triangle in the Argand diagram. Show that $u^2 + v^2 = uv$. *(2 marks)*

ii. Give an example of non-zero complex numbers u and v, so that $0,\ u$ and v form the vertices of an equilateral triangle in the Argand diagram. *(1 mark)*

35. Complex Numbers, EXT2 N2 2019 HSC 16b

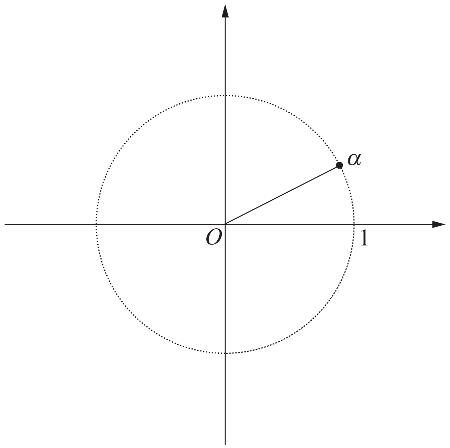
Let $P(z) = z^4 - 2kz^3 + 2k^2z^2 + mz + 1$, where k and m are real numbers.

The roots of $P(z)$ are $\alpha, \overline{\alpha}, \beta, \overline{\beta}$.

It is given that $|\alpha| = 1$ and $|\beta| = 1$.

i. Show that $(\mathbf{Re}(\alpha))^2 + (\mathbf{Re}(\beta))^2 = 1$. ***(3 marks)***

ii. The diagram shows the position of α.



On the diagram, accurately show all possible positions of β. ***(2 marks)***

---

36. Complex Numbers, EXT2 N2 2011 HSC 6c

On an Argand diagram, sketch the region described by the inequality

$$\left| 1 + \frac{1}{z} \right| \leq 1.$$ ***(2 marks)***

---

37. Complex Numbers, EXT2 N2 2016 HSC 16a

i. The complex numbers $z = \cos\theta + i\sin\theta$ and $w = \cos\alpha + i\sin\alpha$, where $-\pi < \theta < \pi$ and $-\pi < \alpha \leq \pi$, satisfy

$$1 + z + w = 0.$$

By considering the real and imaginary parts of $1 + z + w$, or otherwise, show that $1,\ z$ and w form the vertices of an equilateral triangle in the Argand diagram. ***(3 marks)***

ii. Hence, or otherwise, show that if the three non-zero complex numbers $2i,\ z_1$ and z_2 satisfy

$|2i| = |z_1| = |z_2|$ AND $2i + z_1 + z_2 = 0.$

then they form the vertices of an equilateral triangle in the Argand diagram. ***(2 marks)***

---

38. Complex Numbers, EXT2 N2 2021 HSC 16c

Sketch the region of the complex plane defined by $\mathbf{Re}(z) \geq \mathbf{Arg}(z)$ where $\mathbf{Arg}(z)$ is the principal argument of z. ***(3 marks)***

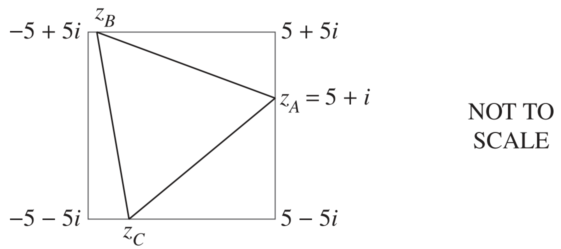
---

39. Complex Numbers, EXT2 N2 2022 HSC 16a

A square in the Argand plane has vertices

$5 + 5i,\ \ 5 - 5i,\ \ -5 - 5i$ and $-5 + 5i$.

The complex numbers $z_A = 5 + i,\ z_B$ and z_C lie on the square and form the vertices of an equilateral triangle, as shown in the diagram.



Find the exact value of the complex number z_B. ***(4 marks)***

Copyright © 2004-23 The State of New South Wales (Board of Studies, Teaching and Educational Standards NSW)

Worked Solutions

1. Complex Numbers, EXT2 N2 2017 HSC 1 MC

$P(x)$ has 8 separate roots.

$\therefore$ Must be of degree at least 8.

Since 1 is also a root,

$\Rightarrow C$

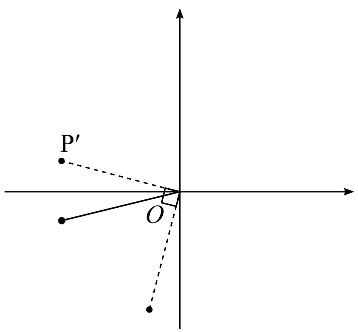
2. Complex Numbers, EXT2 N2 EQ-Bank 1 MC

$e^{i\theta} = \cos\theta + i\sin\theta$

$e^{-i\theta} = \cos(-\theta) + i\sin(-\theta) = \cos\theta - i\sin\theta$ (conjugate)

$e^{i\theta} \Rightarrow P\prime$

$ie^{-i\theta} \Rightarrow$ Rotate $e^{-i\theta}$ $(P\prime)$ 90° anticlockwise.



$\Rightarrow B$

3. Complex Numbers, EXT2 N2 2022 HSC 1 MC

$1 < \text{Re}(z) \leq 3 \quad \Rightarrow \quad$ Eliminate B and D

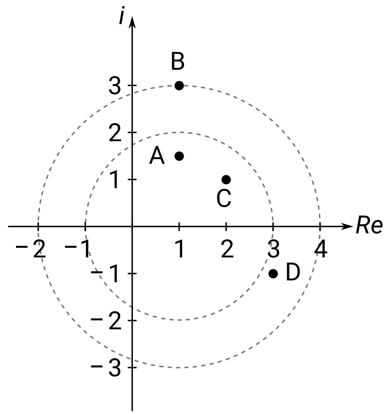
$\dfrac{\pi}{6} \leq \text{Arg}(z) < \dfrac{\pi}{3} \quad \Rightarrow \quad$ Eliminate C

$\Rightarrow A$

4. Complex Numbers, EXT2 N2 2017 HSC 3 MC

$|z-1| = 2$ is a circle with centre $(1, 0)$, radius 2

$|z-1| = 3$ is a circle with centre $(1, 0)$, radius 3



$\therefore$ Only $3-i$ lies within the two circles,

i.e. satisfies $2 < |z-1| < 3$

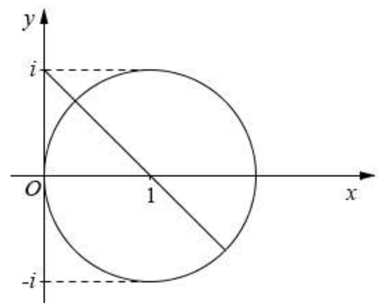
$\Rightarrow D$

5. Complex Numbers, EXT2 N2 2013 HSC 5 MC

$|z-1| = r$ is a circle with centre $(1, 0)$, and radius r.

$\Rightarrow B$

6. Complex Numbers, EXT2 N2 2015 HSC 9 MC



$|z-1| = 1$ is a circle, centre (1,0), radius 1.

Consider the graph:

Distance from $(0, i)$ to the centre $= \sqrt{2}$

$\therefore$ Greatest distance $= \sqrt{2} + \text{radius} = \sqrt{2} + 1$

$\Rightarrow D$

7. Complex Numbers, EXT2 N2 2016 HSC 5 MC

$$\begin{aligned}\frac{1-i}{1+i} &= \frac{(1-i)^2}{(1+i)(1-i)}\\ &= \frac{-2i}{2}\\ &= -i\end{aligned}$$

$\therefore$ Clockwise rotation by $\dfrac{\pi}{2}$.

$\Rightarrow B$

---

8. Complex Numbers, EXT2 N2 2018 HSC 7 MC

$$\begin{aligned}\arg(z) &= \arg(z+1-i)\\ &= \arg(z-(-1+i))\end{aligned}$$

$\Rightarrow$ $\arg(z-(-1+i))$ is the argument of z from $(-1+i)$.

Plot $(-1+i)$ on the argand diagram and then test different positions of z along the solutions for each option.

$\Rightarrow$ D

---

9. Complex Numbers, EXT2 N2 2023 HSC 3 MC

$$z = e^{-\frac{2i\pi}{3}},\ \bar{z} = e^{\frac{2i\pi}{3}}$$

By trial and error:

$$z^2 = e^{-\frac{2\times 2i\pi}{3}} = e^{-\frac{4i\pi}{3}} = e^{\frac{2i\pi}{3}} = \bar{z}$$

$\Rightarrow B$

---

10. Complex Numbers, EXT2 N2 2023 HSC 8 MC

♦♦ Mean mark 34%.

By elimination:

Let $z = x + iy\ (x, y \in \mathbb{R})$

Since shaded area is outside the circle, coefficients of x^2 and y^2 must be positive (eliminate A and C).

Consider option D:

$$\begin{aligned}|z-1| &< 2|z-i|\\ (x-1)^2 + y^2 &< 4(x^2 + (y-1)^2)\\ x^2 - 2x + 1 + y^2 &< 4x^2 + 4y^2 - 8y + 4\\ 0 &< 3x^2 + 2x + 3y^2 - 8y + 3\end{aligned}$$

Circle equation has centre where $x < 0, y > 0$.

$\Rightarrow D$

---

11. Complex Numbers, EXT2 N2 2011 HSC 2b

i. $z = 1 + i\sqrt{3} + \sqrt{3} + i$

$$= \left(1+\sqrt{3}\right) + i\left(1+\sqrt{3}\right)$$

ii. $\arg z = \tan^{-1}\left(\dfrac{1+\sqrt{3}}{1+\sqrt{3}}\right) = \dfrac{\pi}{4}$

$\arg\left(\sqrt{3}+i\right) = \tan^{-1}\left(\dfrac{1}{\sqrt{3}}\right) = \dfrac{\pi}{6}$

Difference $= \dfrac{\pi}{4} - \dfrac{\pi}{6} = \dfrac{\pi}{12}$

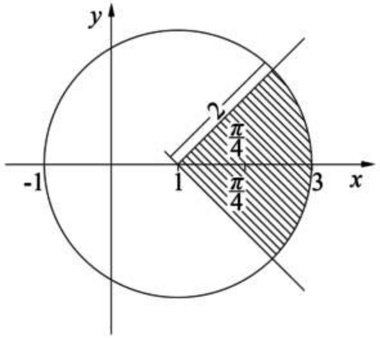
$\Rightarrow$ Opposite angles of a rhombus are equal

$\Rightarrow$ The diagonals of a rhombus bisect the angles

$$\begin{aligned}\therefore \theta &= \pi - 2\times\frac{\pi}{12}\\ &= \frac{5\pi}{6}\quad \text{(angle sum of triangle)}\end{aligned}$$

---

12. Complex Numbers, EXT2 N2 2009 HSC 2d



13. Complex Numbers, EXT2 N2 2010 HSC 2c

Consider $\quad 1 \le |\ z\ | \le 2$

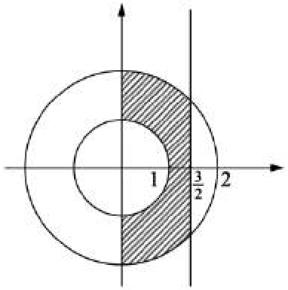
$$1 \le x^2 + y^2 \le 4$$

$$z + \bar{z} = (x + iy) + (x - iy) = 2x$$

Consider $\quad 0 \le z + \bar{z} \le 3$

$$0 \le 2x \le 3$$

$$0 \le x \le \frac{3}{2}$$

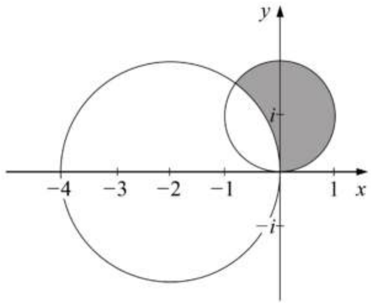


14. Complex Numbers, EXT2 N2 2012 HSC 11b

$|\ z + 2\ | \ge 2$ and $|\ z - i\ | \le 1$

$(x+2)^2 + y^2 \ge 4$

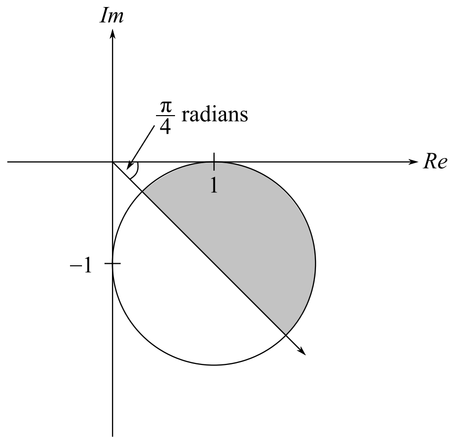
$x^2 + (y-1)^2 \le 1$



15. Complex Numbers, EXT2 N2 2017 HSC 11c

$|z - 1 + i| = 1$ is a circle with centre $(1, -1)$ and radius 1.

Shaded area: $-\frac{\pi}{4} \le \arg(z) \le 0 \ \cap \ |z - 1 + i| \le 1$



16. Complex Numbers, EXT2 N2 2018 HSC 11d

i. $w = iu$

$$= i(5+2i)$$

$$= -2+5i$$

ii. $v = u + w$

$$= 5+2i+(-2+5i)$$

$$= 3+7i$$

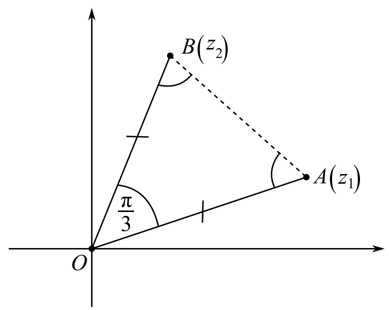
iii. $\arg\left(\dfrac{w}{v}\right) = \arg(w) - \arg(v)$

$$= \frac{\pi}{4} \quad \text{(diagonal of square bisects corner)}$$

---

17. Complex Numbers, EXT2 N2 2020 HSC 14a

i.



Let $z_1 = r(\cos\theta + i\sin\theta)$

$$z_2 = e^{i\frac{\pi}{3}} z_1$$

$$= r\left(\cos\left(\theta + \frac{\pi}{3}\right) + i\sin\left(\theta + \frac{\pi}{3}\right)\right)$$

$$|z_1| = |z_2| \Rightarrow OA = OB$$

$$\angle AOB = \frac{\pi}{3} \quad \left(z_2 \text{ is a } \frac{\pi}{3} \text{ anti-clockwise rotation of } z_1\right)$$

$$\Rightarrow \angle OBA = \angle BAO = \frac{\pi}{3} \quad \text{(angles opposite equal sides)}$$

$$\therefore\ OAB \text{ is equilateral}$$

ii.

$$z_1 = z_2 e^{i\frac{\pi}{3}}$$

$$\frac{z_1}{z_2} = e^{i\frac{\pi}{3}}$$

$$\left(\frac{z_1}{z_2}\right)^3 = e^{\left(3 \times i\frac{\pi}{3}\right)} \quad \text{(by De Moivre)}$$

$$\frac{z_1^3}{z_2^3} = e^{i\pi}$$

$$z_1^3 = -z_2^3$$

$$z_1^3 + z_2^3 = 0$$

$$(z_1+z_2)\left(z_1^2 - z_1z_2 + z_2^2\right) = 0$$

$$z_1^2 - z_1z_2 + z_2^2 = 0$$

$$z_1^2 + z_2^2 = z_1z_2$$

COMMENT: The identity to prove suggests the addition of 2 cubes is a possible strategy.

---

18. Complex Numbers, EXT2 N2 2007 HSC 2c

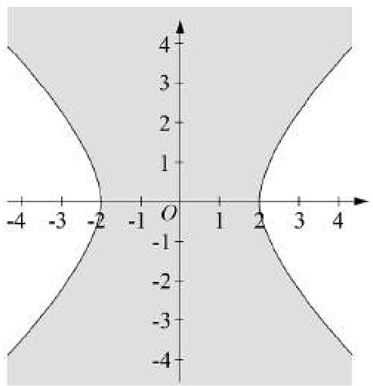
$$\frac{1}{z}+\frac{1}{\bar{z}}=\frac{1}{x+iy}+\frac{1}{x-iy}$$
$$=\frac{x-iy+x+iy}{x^2+y^2}$$
$$=\frac{2x}{x^2+y^2}$$

$$\frac{2x}{x^2+y^2}=1$$
$$x^2+y^2=2x$$
$$x^2-2x+1+y^2=1$$
$$(x-1)^2+y^2=1$$

$\therefore$ Locus is a circle, centre $(1,0)$, radius 1, excluding the point $(0,0)$.

19. Complex Numbers, EXT2 N2 2013 HSC 11e

$$z^2+\bar{z}^2\le 8$$
$$(x+iy)^2+(x-iy)^2\le 8$$
$$x^2+2ixy-y^2+x^2-2ixy-y^2\le 8$$
$$2x^2-2y^2\le 8$$
$$x^2-y^2\le 4$$

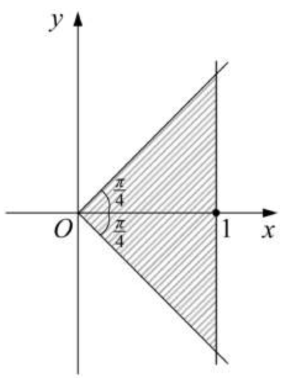


20. Complex Numbers, EXT2 N2 2014 HSC 11c

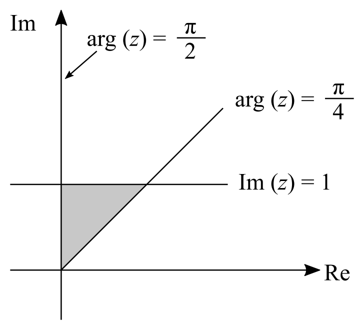
Let $z=x+iy$

$$|x+iy|\le|x-2+iy|$$
$$\sqrt{x^2+y^2}\le\sqrt{(x-2)^2+y^2}$$
$$x^2+y^2\le x^2-4x+4+y^2$$
$$4x-4\le 0$$
$$x\le 1$$

$$-\frac{\pi}{4}\le \arg z\le\frac{\pi}{4}$$



21. Complex Numbers, EXT2 N2 2019 HSC 12a

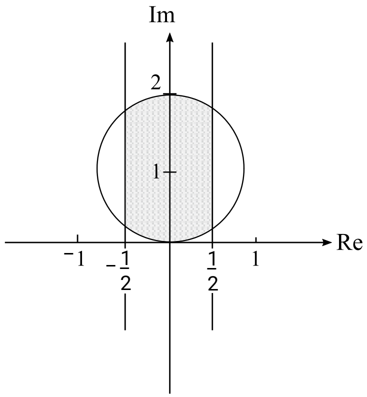


Shaded Area: $\frac{\pi}{4}\le\arg(z)\le\frac{\pi}{2}$ and $\mathrm{Im}(z)\le 1$

22. Complex Numbers, EXT2 N2 2004 HSC 2c

$$|z+\bar{z}| \leq 1$$
$$|x+iy+x-iy| \leq 1$$
$$|2x| \leq 1$$
$$|x| \leq \frac{1}{2}$$

$|z-i| \leq 1 \;\Rightarrow\;$ Circle, radius $=1$, centre $(0,1)$

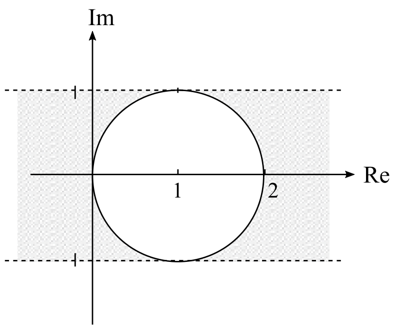


23. Complex Numbers, EXT2 N2 2005 HSC 2c

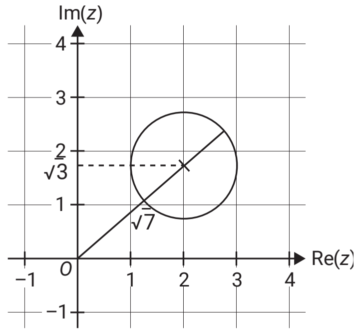
$$|z-\bar{z}| < 2$$
$$|x+iy-(x-iy)| < 2$$
$$|2iy| < 2$$
$$|y| < 1$$

$|z-1| = 1 \;\Rightarrow\;$ Circle, radius $=1$, centre $(1,0)$

$\therefore$ Graph: $|z-\bar{z}| < 2 \,\cap\, |z-1| \geq 1$



24. Complex Numbers, EXT2 N2 2021 SPEC2 5



Centre of circle at $\left(2, \sqrt{3}\right)$

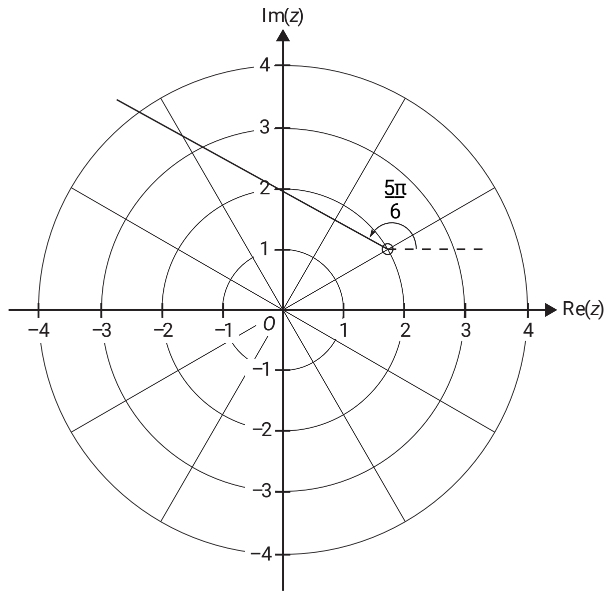
Radius $=1$

By Pythagoras, line from origin to the centre of the circle

$$d = \sqrt{2^2 + \sqrt{3}^2} = \sqrt{7}$$

$$\therefore |z|_{\max} = \sqrt{7}+1$$

25. Complex Numbers, EXT2 N2 SM-Bank 11



26. Complex Numbers, EXT2 N2 2007 HSC 2d

i. Since ΔORQ is equilateral, each angle is $\dfrac{\pi}{3}$ radians.

From $R(a)$:

$Q(z_2)$ is an anticlockwise rotation through $\dfrac{\pi}{3}$.

$$\therefore z_2 = a\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) = \omega a.$$

ii. Solution 1

Similarly, $a = z_1\omega$

$$\begin{aligned} z_1 &= \frac{a}{\omega} \\ : z_1 z_2 &= \frac{a}{\omega} \times \omega a \\ &= a^2 \end{aligned}$$

Solution 2

$P(z_1)$ is a clockwise rotation of $R(a)$ through $\dfrac{\pi}{3}$.

$$\therefore z_1 = \overline{\omega} a.$$

$$\begin{aligned} \therefore z_1 z_2 &= \overline{\omega} a \times \omega a \\ &= a^2\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right) \times \left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) \\ &= a^2\left(\cos^2\frac{\pi}{3} + \sin^2\frac{\pi}{3}\right) \\ &= a^2 \end{aligned}$$

iii. $z^2 - az + a^2 = 0$

Let the roots be α and β.

$$\begin{aligned} \alpha + \beta &= -\frac{b}{a} = a \\ \alpha\beta &= \frac{c}{a} = a^2 \end{aligned}$$

$$\begin{aligned} z_1 z_2 &= a^2 \quad \text{(part (ii))} \\ z_1 + z_2 &= \overline{\omega} a + \omega a \\ &= \left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3} + \cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right) a \\ &= 2\cos\frac{\pi}{3} \times a \\ &= 2 \times \frac{1}{2} \times a \\ &= a \end{aligned}$$

$\therefore z_1$ and z_2 are the roots of $z^2 - az + a^2 = 0$.

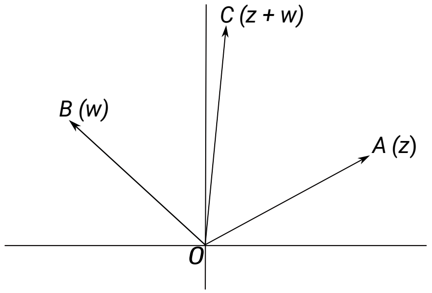
27. Complex Numbers, EXT2 N2 2023 14a

i. $z = e^{\frac{i\pi}{6}} = \cos\frac{\pi}{6} + i\sin\frac{\pi}{6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$

$w = e^{\frac{3i\pi}{4}} = \cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4} = -\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$

$$\begin{aligned}
|z+w|^2 &= \left|\frac{\sqrt{3}}{2} + \frac{1}{2}i - \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right| \\
&= \left|\left(\frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}\right) + \left(\frac{1}{2} + \frac{1}{\sqrt{2}}\right)i\right| \\
&= \left|\frac{\sqrt{6}-2}{2\sqrt{2}} + \frac{\sqrt{2}+2}{2\sqrt{2}}i\right| \\
&= \frac{(\sqrt{6}-2)^2 + (\sqrt{2}+2)^2}{(2\sqrt{2})^2} \\
&= \frac{6 - 4\sqrt{6} + 4 + 2 + 4\sqrt{2} + 4}{8} \\
&= \frac{16 - 4\sqrt{6} + 4\sqrt{2}}{8} \\
&= \frac{4 - \sqrt{6} + \sqrt{2}}{2}
\end{aligned}$$

ii.



$\angle AOB = \arg(w) - \arg(z) = \frac{3\pi}{4} - \frac{\pi}{6} = \frac{7\pi}{12}$

$|z| = |w| = 1 \Rightarrow AOBC$ is a rhombus.

$\overrightarrow{OC}$ is a diagonal of rhombus $AOBC$

$\Rightarrow \overrightarrow{OC}$ bisects $\angle AOB$

$\therefore \angle AOC = \frac{1}{2} \times \frac{7\pi}{12} = \frac{7\pi}{24}$

iii. In $\triangle AOC$:

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \overrightarrow{OB}$

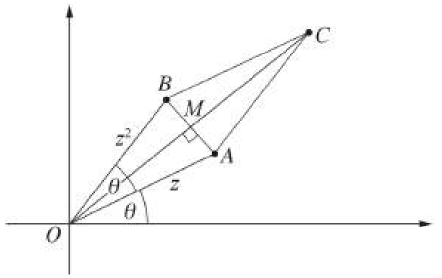
$\Rightarrow \overrightarrow{OB}$ is represented by w.

Using the cos rule in $\triangle AOC$:

$$\begin{aligned}
\cos\frac{7\pi}{24} &= \frac{|z|^2 + |z+w|^2 - |w|^2}{2|z||z+w|} \\
&= \frac{1 + \frac{4-\sqrt{6}+\sqrt{2}}{2} - 1}{2 \times 1\sqrt{\frac{4-\sqrt{6}+\sqrt{2}}{2}}} \\
&= \frac{\sqrt{\frac{4-\sqrt{6}+\sqrt{2}}{2}} \times 2}{2 \times 2} \\
&= \frac{\sqrt{4\left(\frac{4-\sqrt{6}+\sqrt{2}}{2}\right)}}{4} \\
&= \frac{8 - 2\sqrt{6} + 2\sqrt{2}}{4}
\end{aligned}$$

28. Complex Numbers, EXT2 N2 2010 HSC 2d

i. $z = \cos\theta + i\sin\theta$



$|OA| = |z| = 1$

$|OB| = |z^2| = |z|^2 = 1$

$\therefore$ $OACB$ is a parallelogram with a pair of adjacent sides equal.

$\therefore$ $OACB$ is a rhombus.

ii. $\arg z^2 = 2\arg z = 2\theta$

$\angle BOA = 2\theta - \theta = \theta$

Since $OACB$ is a rhombus then CO bisects $\angle BOA$

$\therefore \angle COA = \dfrac{\theta}{2}$

$\therefore$ $\arg(z+z^2) = \theta + \dfrac{\theta}{2} = \dfrac{3\theta}{2}$

iii. $OC = |z+z^2|$

Join AB so that it meets OC at M

$AB \perp OC$, and $OM = OC$ (diagonals of a rhombus)

In ΔOAM:

$$\cos\frac{\theta}{2} = \frac{OM}{OA}$$
$$= OM$$
$$\therefore OC = 2 \times OM$$
$$= 2\cos\frac{\theta}{2}$$

♦♦ *Mean mark part (iii) 26%.*

iv. $z + z^2 = \cos\theta + i\sin\theta + (\cos\theta + i\sin\theta)^2$

$= \cos\theta + \cos 2\theta + i(\sin\theta + \sin 2\theta)$ (De Moivre)

$\therefore$ $\text{Re}(z+z^2) = \cos\theta + \cos 2\theta$

Using parts (ii) and (iii),

♦♦ *Mean mark part (iv) 44%.*

$$z + z^2 = 2\cos\frac{\theta}{2}\left(\cos\frac{3\theta}{2} + i\sin\frac{3\theta}{2}\right)$$

Equating real parts:

$$\cos\theta + \cos 2\theta = 2\cos\frac{\theta}{2}\cos\frac{3\theta}{2}$$

---

29. Complex Numbers, EXT2 N2 2011 HSC 4a

i.

$$|z-a|^2 - |z-b|^2 = 1$$
$$|(x-a)+iy|^2 - |(x-b)+iy|^2 = 1$$
$$(x-a)^2 + y^2 - \left((x-b)^2 + y^2\right) = 1$$
$$(x-a)^2 - (x-b)^2 = 1$$
$$(x-a-(x-b))(x-a+x-b) = 1$$
$$(b-a)(2x-a-b) = 1$$

$$2x-a-b = \frac{1}{b-a}$$
$$2x = a+b+\frac{1}{b-a}$$
$$\therefore x = \frac{a+b}{2} + \frac{1}{2(b-a)}$$

ii. The locus is the vertical line

$$x = \frac{a+b}{2} + \frac{1}{2(b-a)}.$$

♦ *Mean mark part (ii) 44%.*

---

30. Complex Numbers, EXT2 N2 2012 HSC 12d

i. $\overrightarrow{A_1P} = z - u_1$

$\overrightarrow{A_1B_1} = w_1 - u_1$

$B_1A_1 \perp A_1P$ and $\left|\overrightarrow{A_1P}\right| = \left|\overrightarrow{A_1B_1}\right|$

$\overrightarrow{A_1B_1}$ is an anticlockwise rotation of $\overrightarrow{A_1P}$ through $90°$

$\therefore w_1 - u_1 = i(z - u_1)$

$\therefore w_1 = u_1 + i(z - u_1)$

ii. $\overrightarrow{A_2B_2} = w_2 - u_2$

$\overrightarrow{A_2P} = z - u_2$

$A_2B_2 \perp A_2P$ and $\left|\overrightarrow{A_2B_2}\right| = \left|\overrightarrow{A_2P}\right|$

$\overrightarrow{A_2P}$ is an anticlockwise rotation of $\overrightarrow{A_2B_2}$ through $90°$

$$\begin{aligned} z - u_2 &= i(w_2 - u_2) \\ iw_2 &= z - u_2 + iu_2 \\ -w_2 &= iz - iu_2 - u_2 \\ \therefore\ w_2 &= u_2 + i(u_2 - z) \end{aligned}$$

$\therefore$ The midpoint of B_1B_2 is $\dfrac{w_1 + w_2}{2}$

$$= \frac{1}{2}[u_1 + i(z - u_1) + u_2 + i(u_2 - z)]$$

$$= \frac{1}{2}[u_1 + u_2 + i(u_2 - u_1)]$$

$$= \frac{u_1 + u_2}{2} + \frac{u_2 - u_1}{2}i \quad \text{(which is a fixed point)}$$

31. Complex Numbers, EXT2 N2 2020 SPEC2 2

a. Let $z = x + iy$

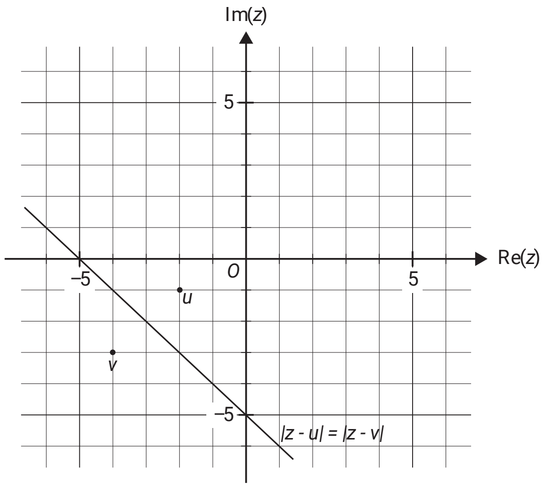
$z - u = x + 2 + iy + i$

$z - v = x + 4 + iy + 3i$

$|z-u| = |z-v|$

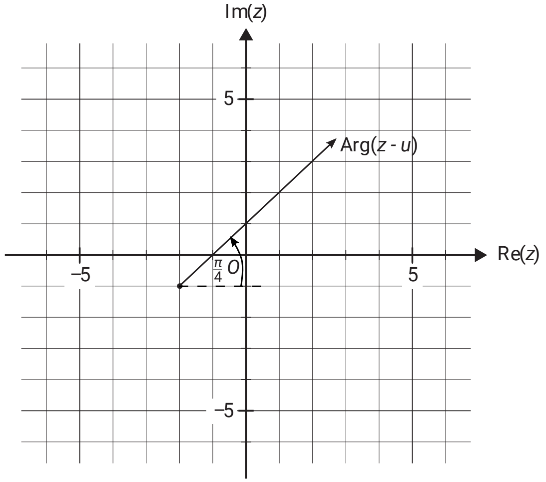
$$\begin{aligned} (x+2)^2 + (y+1)^2 &= (x+4)^2 + (y+3)^2 \\ x^2 + 4x + 4 + y^2 + 2y + 1 &= x^2 + 8x + 16 + y^2 + 6y + 9 \\ -4y &= 4x + 20 \\ y &= -x - 5 \end{aligned}$$

b.



c. $|z-u| = |z-v|$ is the graph of the perpendicular bisector of the line joining u and v.

d.i.



d.ii. $\text{Arg}(z-u) = \dfrac{\pi}{4} \Rightarrow$ gradient $= 1$, y-intercept at $(0, 1)$

$\therefore f: (-2, \infty) \to \mathbb{R},\ f(x) = x + 1$

32. Complex Numbers, EXT2 N2 2013 HSC 15a

$$A = \frac{1}{2} ab\sin C$$
$$= \frac{1}{2}|z||w|\ \sin(\theta - \phi)$$

♦♦ *Mean mark 29%.*
STRATEGY: The angles shown in the graphic should alert students that the mod-arg approach is likely to be easier than the $x + iy$ form.

$$z\overline{w} - w\overline{z} = |z|\text{cis}\ \theta \cdot |w|\text{cis}(-\phi) - |w|\text{cis}\ \phi \cdot |z|\text{cis}(-\theta)$$
$$= |z||w|(\text{cis}\ \theta\ \text{cis}(-\phi) - \text{cis}\ \phi\ \text{cis}(-\theta))$$
$$= |z||w|(\text{cis}(\theta - \phi) - \text{cis}(\phi - \theta))$$

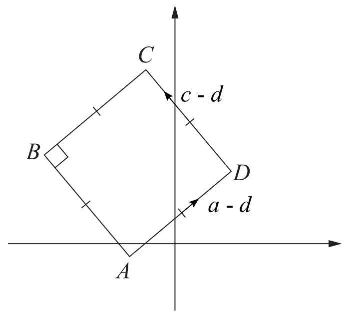
Since $\cos(\phi - \theta) = \cos(\theta - \phi)$, and

$\sin(\phi - \theta) = -\sin(\theta - \phi)$

$\Rightarrow \text{cis}(\theta - \phi) - \text{cis}(\phi - \theta) = 2i\sin(\theta - \phi)$

$$\therefore z\overline{w} - w\overline{z} = 2i|z||w|\sin(\theta - \phi)$$
$$= 4i\left(\frac{1}{2}|z||w|\ \sin(\theta - \phi)\right)$$
$$= 4iA \quad \ldots \text{ as required}$$

33. Complex Numbers, EXT2 N2 2017 HSC 13e



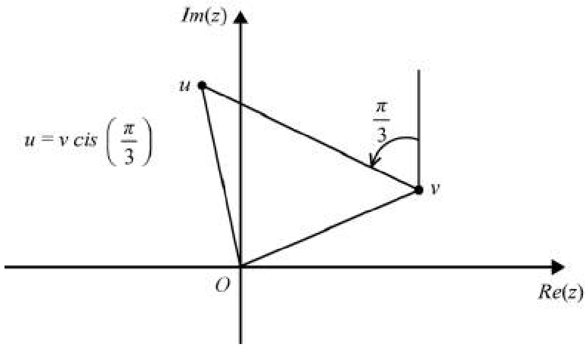
$$c - d = i(d - a) \quad (\text{rotation of } \frac{\pi}{2})$$
$$\therefore c = d + ia$$
$$= (1 + i)d - ia \ \ldots \text{ as required.}$$

♦ *Mean mark 49%.*

34. Complex Numbers, EXT2 N2 2016 HSC 16b

i. $0, u, v$ are vertices of an equilateral triangle.

♦♦ *Mean mark 23%.*



$$\Rightarrow u = v\text{cis}\left(\pm\frac{\pi}{3}\right)$$

$$u^3 = v^3\text{cis}(\pm\pi)$$

$$u^3 = -v^3$$

$$u^3 + v^3 = 0$$

$$(u+v)\left(u^2 - uv + v^2\right) = 0$$

Since $u \neq v$,

$$u^2 - uv + v^2 = 0$$

$$\therefore u^2 + v^2 = uv$$

ii. Let $v = 1$,

♦♦ *Mean mark 33%.*

$$\therefore u = \text{cis}\left(\frac{\pi}{3}\right)$$

$$= \cos\frac{\pi}{3} + i\sin\frac{\pi}{3}$$

$$= \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

35. Complex Numbers, EXT2 N2 2019 HSC 16b

i. $P(z) = z^4 - 2kz^3 + 2k^2z^2 + mz + 1, \quad k, m \in \mathbb{R}$

Roots: $\alpha, \overline{\alpha}, \beta, \overline{\beta}$ and $|\alpha| = 1, |\beta| = 1$

Show $(\text{Re}(\alpha))^2 + (\text{Re}(\beta))^2 = 1$

♦♦ *Mean mark part (i) 26%.*

$$\alpha + \overline{\alpha} + \beta + \overline{\beta} = 2k$$

$$2\text{Re}(\alpha) + 2\text{Re}(\beta) = 2k$$

$$\text{Re}(\alpha) + \text{Re}(\beta) = k$$

$$\alpha\overline{\alpha} + \alpha\beta + \alpha\overline{\beta} + \overline{\alpha}\beta + \overline{\alpha}\overline{\beta} + \beta\overline{\beta} = 2k^2$$

$$|\alpha|^2 + \alpha\left(\beta + \overline{\beta}\right) + \overline{\alpha}\left(\beta + \overline{\beta}\right) + |\beta|^2 = 2k^2$$

$$1 + (\alpha + \overline{\alpha})\left(\beta + \overline{\beta}\right) + 1 = 2k^2$$

$$2 + 2\text{Re}(\alpha) \cdot 2\text{Re}(\beta) = 2(\text{Re}(\alpha) + \text{Re}(\beta))^2$$

$$2 + 4\text{Re}(\alpha)\text{Re}(\beta) = 2\text{Re}(\alpha)^2 + 4\text{Re}(\alpha)\text{Re}(\beta) + 2\text{Re}(\beta)^2$$

$$2 = 2\left(\text{Re}(\alpha)^2 + \text{Re}(\beta)^2\right)$$

$$\therefore 1 = \text{Re}(\alpha)^2 + \text{Re}(\beta)^2$$

ii. $|\alpha| = |\beta|$ (given)

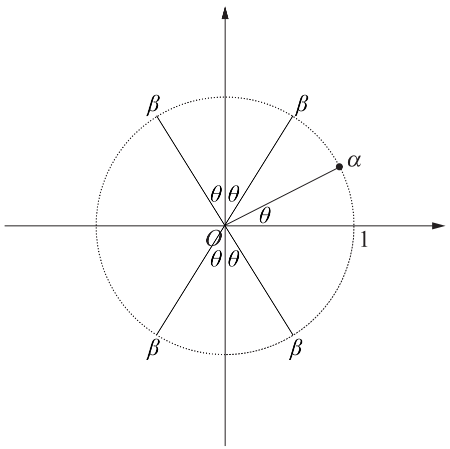
$\text{Re}(\alpha)^2 + \text{Re}(\beta)^2 = 1$ (see part (i))

$\text{Re}(\alpha)^2 + \text{Im}(\alpha)^2 = 1$ $(|\alpha| = 1)$

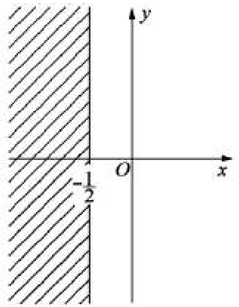
$\Rightarrow \text{Re}(\beta)^2 = \text{Im}(\alpha)^2$

$\text{Re}(\beta) = \pm\,\text{Im}(\alpha)$

♦♦♦ *Mean mark part (ii) 10%.*



36. Complex Numbers, EXT2 N2 2011 HSC 6c



$$\left|1+\frac{1}{z}\right| \le 1$$
$$\left|\frac{z+1}{z}\right| \le 1$$
$$\frac{|z+1|}{|z|} \le 1$$
$$|z+1| \le |z|$$
$$\sqrt{(x+1)^2+y^2} \le \sqrt{x^2+y^2}$$
$$(x+1)^2+y^2 \le x^2+y^2$$
$$x^2+2x+1 \le x^2$$
$$\therefore x \le -\frac{1}{2}$$

♦♦♦ *Mean mark 18%.*
MARKER'S COMMENT: Substituting $z=x+iy$ immediately was common and caused major algebraic problems.

---

37. Complex Numbers, EXT2 N2 2016 HSC 16a

♦♦♦ *Mean mark 25%.*
STRATEGY: A clear graphical image simplifies both parts of this question significantly.

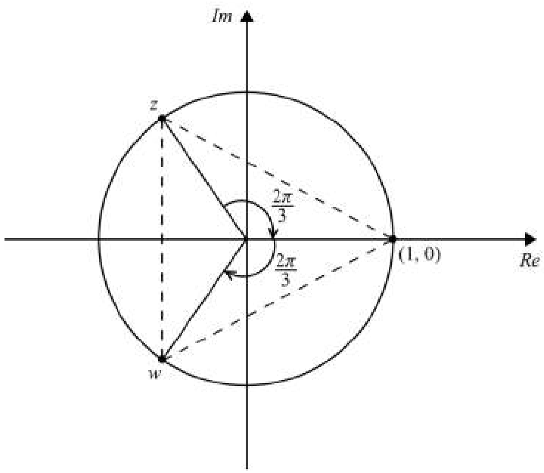
i. $z = \cos\theta + i\sin\theta,\ |z| = 1$

$w = \cos\alpha + i\sin\alpha,\ |w| = 1$

Since $1+z+w=0$

$$\text{Im}(1+z+w) = 0$$
$$\sin\theta + \sin\alpha = 0$$
$$\sin\theta = -\sin\alpha$$
$$\therefore \theta = -\alpha \ldots (1)$$

$$\text{Re}(1+z+w) = 0$$
$$1+\cos\theta+\cos\alpha = 0$$
$$2\cos\theta = -1 \quad (\text{Since } \cos\alpha = \cos(-\theta) = \cos\theta)$$
$$\cos\theta = -\frac{1}{2}$$
$$\therefore \theta = \frac{2\pi}{3}$$
$$\alpha = -\frac{2\pi}{3}$$



$\Rightarrow$ All points are on the unit circle separated by $\frac{2\pi}{3}$ radians.

$\therefore$ They are vertices of an equilateral triangle.

ii. $|2i| = 2$

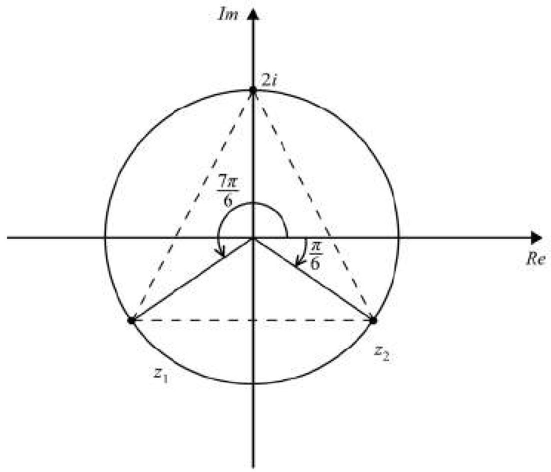
Let $z_1 = 2(\cos\theta + i\sin\theta),\ |z_1| = 2$

$z_2 = 2(\cos\alpha + i\sin\alpha), \quad |z_2| = 2$

$$\text{Re}(2i + z_1 + z_2) = 0$$
$$2(\cos\theta + \cos\alpha) = 0$$
$$\therefore \cos\theta = -\cos\alpha$$
$$\therefore \theta = \pi - \alpha$$

♦♦♦ *Mean mark 5%.*

$$\text{Im}(2i + z_1 + z_2) = 0$$
$$2(1 + \sin\theta + \sin\alpha) = 0$$
$$\sin\theta + \sin\alpha = -1$$
$$2\sin\theta = -1 \quad (\text{Since } \sin\alpha = \sin(\pi - \theta) = \sin\theta)$$
$$\sin\theta = -\frac{1}{2}$$
$$\therefore \theta = \frac{7\pi}{6}$$
$$\therefore \alpha = -\frac{\pi}{6}$$



$\Rightarrow$ All points are on the 2 unit circle separated by $\dfrac{2\pi}{3}$ radians.

$\therefore 2i, z_1$ and z_2 are vertices of an equilateral triangle.

---

38. Complex Numbers, EXT2 N2 2021 HSC 16c

Find region where $\text{Re}(z) \geq \text{Arg}(z)$.

In quadrant 1: $x \geq 0,\ y \geq 0,\ 0 \leq \text{Arg}(z) \leq \dfrac{\pi}{2}$

$$x \geq \tan^{-1}\left(\frac{y}{x}\right)$$
$$x\tan(x) \geq y$$

♦♦♦ *Mean mark 15%.*

If $x \geq \dfrac{\pi}{2},\ x \geq \tan^{-1}\left(\dfrac{y}{x}\right)$ as $\tan^{-1}\left(\dfrac{y}{x}\right) < \dfrac{\pi}{2}$

$\text{Arg}(0)$ is not defined.

In quadrant 2: $x \leq 0,\ y \leq 0,\ \dfrac{\pi}{2} \leq \text{Arg}(z) \leq \pi$

$\text{Re}(z) < \text{Arg}(z) \Rightarrow$ no points satisfy

In quadrant 3: $x \leq 0,\ y \leq 0,\ -\pi \leq \text{Arg}(z) \leq -\dfrac{\pi}{2}$

$$\text{Arg}(z) = -\pi + \tan^{-1}\left(\frac{y}{x}\right)$$
$$x \geq -\pi + \tan^{-1}\left(\frac{y}{x}\right)$$
$$\tan^{-1}\left(\frac{y}{x}\right) \leq \pi + x$$
$$\frac{y}{x} \leq \tan(\pi + x)$$
$$y \geq x\tan(\pi + x)$$

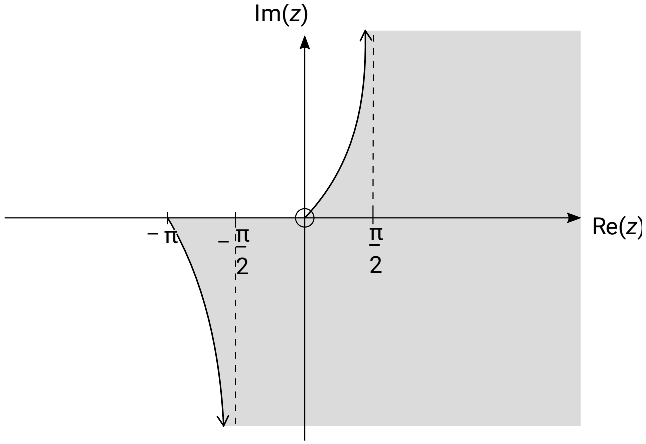
In the domain $-\dfrac{\pi}{2} \leq x \leq 0,$

$\text{Arg}(z) = -\pi + \tan^{-1}\left(\dfrac{y}{x}\right) < -\dfrac{\pi}{2} \quad \left(\text{since } \tan^{-1}\left(\dfrac{y}{x}\right) < \dfrac{\pi}{2}\right)$

$\Rightarrow$ all points satisfy for $-\dfrac{\pi}{2} \leq x \leq 0$

In quadrant 4: $x \geq 0,\ y \leq 0,\ -\dfrac{\pi}{2} \leq \text{Arg}(z) \leq 0$

$\text{Re}(z) > \text{Arg}(z) \Rightarrow$ all points satisfy



39. Complex Numbers, EXT2 N2 2022 HSC 16a

$$z_A = 5 + i, \quad z_B = b + 5i, \quad z_C = c - 5i$$
$$z_B - z_A = (b-5) + 4i$$
$$z_C - z_A = (c-5) - 6i$$

♦♦♦ Mean mark 21%.

Internal angles of equilateral triangle $= \dfrac{\pi}{3}$:

$\Rightarrow$ $(z_B - z_A)$ **is an anti-clockwise rotation of** $(z_C - z_A)$ **by** $\dfrac{\pi}{3}$

$$e^{i\frac{\pi}{3}}(z_B - z_A) = z_C - z_A$$
$$\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)((b-5) + 4i) = (c-5) - 6i$$
$$\frac{b-5}{2} + 2i + \frac{(b-5)\sqrt{3}}{2}i - 2\sqrt{3} = (c-5) - 6i$$
$$\frac{b-5-4\sqrt{3}}{2} + i\left(\frac{4 + (b-5)\sqrt{3}}{2}\right) = (c-5) - 6i$$

Equating imaginary parts:

$$\frac{4 + (b-5)\sqrt{3}}{2} = -6$$
$$(b-5)\sqrt{3} = -16$$
$$b - 5 = -\frac{16}{\sqrt{3}}$$
$$b = 5 - \frac{16}{\sqrt{3}}$$

$$\therefore z_B = \left(5 - \frac{16}{\sqrt{3}}\right) + 5i$$

Copyright © 2016-2024 M2 Mathematics Pty Ltd (SmarterMaths.com.au)